

Stony Brook University
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Spring 2019

ASSIGNMENT 5: Choice under uncertainty

Due: Monday April 22, in class.

Instructions: Show all your work or points will be deducted.

1. The *certainty equivalent* of a lottery is the amount of money you would have to be given with certainty to be just as well-off with that lottery. Suppose that your (von Neumann-Morgenstern) expected utility function over lotteries that give you an amount of money x_A if event A happens and an amount x_{-A} if event A does not happen is $U(x_A, x_{-A}, \pi) = \pi \sqrt{x_A} + (1 - \pi) \sqrt{x_{-A}}$, where π is the probability that event A happens and $1 - \pi$ is the probability that event A does not happen.
 - (a) If $\pi = 1/2$, calculate the utility of a lottery that gives you \$10,000 if event A happens and \$100 if event A doesn't happen.
 - (b) If you were sure to receive \$4,900, what would your utility be?
 - (c) Given this utility function and $\pi = 1/2$, write a general formula for the certainty equivalent of a lottery that gives you an amount of money x_A if event A happens and an amount x_{-A} if event A does not happen.
 - (d) Calculate the certainty equivalent of receiving \$10,000 if event A happens and \$100 if event A does not happen.
2. Mr. P is a risk averter who tries to maximize the expected value of $\sqrt{W}$, where¹ W is his wealth. Mr. P has \$50,000 in safe assets and he also owns a house that is located in an area where there are lots of forest fires. If his house burns down, the remains of his house and the lot it is built on would be worth only \$40,000, giving him a total wealth of \$90,000. If his home doesn't burn, it will be worth \$200,000 and his total wealth will be \$250,000. The probability that his home will burn down is 0.01.
 - (a) Calculate his expected utility if he doesn't buy fire insurance.
 - (b) Calculate the certainty equivalent of the lottery he faces if he doesn't buy fire insurance.
 - (c) Suppose that he can buy insurance at a price of \$1 per \$100 of insurance. For example, if he buys \$100,000 worth of insurance, he will pay \$1,000 to the company no matter what happens, but if his house burns, he will also receive \$100,000 from the company. If Mr. P buys \$160,000 worth of insurance, show that his total wealth is independent of whether there is a fire or not, i.e. wealth is constant across states.
 - (d) If Mr. P buys full insurance, what is the certainty equivalent of his wealth? What is his expected utility?
3. Mr. Willy owns a small chocolate factory, located close to a river that occasionally floods in the spring, with disastrous consequences. Next summer, Willy plans to sell the factory and retire. The only income he will have is the proceeds of the sale of his factory. If there is no flood, the factory will be worth \$500,000. If there is a flood, then what is left of the factory will be worth only

¹ $\sqrt{W}$ is the Bernoulli Utility function.

\$50,000. Willy can buy flood insurance at a cost of \$0.10 for each \$1 worth of coverage. Willy thinks that the probability that there will be a flood this spring is $1/10$. Let c_F denote the contingent commodity dollars if there is a flood and c_{NF} denote dollars if there is no flood. Willy's (von Neumann-Morgenstern) expected utility function is² $U(c_F, c_{NF}) = 0.1 \log(c_F) + 0.9 \log(c_{NF})$.

- (a) If he buys no insurance, then in each contingency, what will be Willy's contingent level of consumption?
- (b) To buy insurance that pays him $\$x$ in the case of a flood, Willy must pay an insurance premium of $.1x$. Such an insurance contract will pay an amount $\$x$ in the event of a flood. Suppose that Willy has contracted for insurance that pays $\$x$ in the event of a flood. After paying for his premium, what will be Willy's contingent consumption level in the event of a flood? What will be his contingent consumption level if there is no flood?
- (c) Eliminate x from the two equations obtained in the previous item to obtain Willy's *state-contingent budget constraint*.
- (d) What is Willy's marginal rate of substitution between the two contingent commodities (dollars if there is no flood and dollars if there is a flood)?
- (e) Find Willy's optimal bundle of contingent commodities.
- (f) Given your answer in the previous item, compute the optimal amount x of insurance that Willy will buy and the insurance premium that he will have to pay.

²In this case, the Bernoulli utility function is $\log(c)$.